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Cybersecurity disclosure and digital maturity as strategic drivers of banking performance: Evidence from Europe

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ABSTRACT

This study examines whether cybersecurity disclosure (CSD) is associated with bank profitability and whether national digital maturity conditions this relationship across European banking systems. Prior research largely focuses on valuation effects, providing limited evidence on the operational governance implications of cybersecurity transparency. Using 8890 bank-year observations from 889 banks across 21 European countries (2014–2023), and employing a panel-data framework with fixed effects and complementary robustness and endogeneity checks, the study develops a dual CSD measure capturing disclosure presence and disclosure depth. Results show that CSD is positively associated with profitability. Disclosure depth demonstrates stronger explanatory power than disclosure presence, suggesting the importance of substantive cybersecurity reporting. National digital maturity as a critical institutional boundary condition shapes the profitability association of CSD, particularly among large banks. Overall, the findings suggest that CSD operates primarily as an operational governance mechanism whose effectiveness depends on supportive digital-institutional environments within European banking systems.

1. Introduction

The rapid digital transformation of financial services has fundamentally reshaped how banks operate, compete, and manage risk. European banking institutions increasingly rely on interconnected information systems, cloud infrastructures, artificial intelligence, and open-banking platforms to deliver financial services more efficiently. While these technologies improve service integration and operational efficiency, they simultaneously expand banks' exposure to cybersecurity threats. Recent cyber incidents—including ransomware attacks, large-scale data breaches, and supply-chain intrusions—have disrupted banking operations, triggered regulatory penalties, and increased supervisory scrutiny across Europe (Allahham et al., 2024b; Cele & Kwenda, 2025; ENISA, 2023). In response, CSD has emerged as an important governance mechanism through which banks communicate how they identify, monitor, and manage cyber risks.

Despite growing attention to cybersecurity transparency, its economic consequences remain debated. Some studies emphasise disclosure-related legitimacy and accountability effects, while others warn that it may reveal vulnerabilities or increase compliance burdens (Barry et al., 2022; Hasani et al., 2023; Héroux & Fortin, 2025; Masoud & Al-Utaibi, 2022). Importantly, most prior research has focused on investor perceptions and market valuation effects rather than realised operational performance. This distinction is particularly relevant for banking institutions, where cyber incidents directly affect service continuity, compliance costs, and risk exposure. Profitability therefore offers a more direct indicator of whether cybersecurity governance practices create measurable organisational value through internal efficiency and risk-management channels (Elsayed et al., 2024; Gao et al., 2020).

Within Europe, the effectiveness of cybersecurity governance is closely tied to the broader digital environment in which banks operate. European financial institutions function within highly integrated digital ecosystems shaped by regulatory initiatives such as PSD2, digital

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identity frameworks, and expanding e-government infrastructures (Demertzis & Wolff, 2020). Prior digital transformation research suggests that firm-level governance outcomes depend partly on the maturity of national digital ecosystems (Abdullah and Almaqtari, 2024; Laitsou et al., 2025; Imran et al., 2022). National digital maturity, captured by the Digital Economy and Society Index (DESI), reflects structural digital capabilities including connectivity, cybersecurity infrastructure, digital skills, business digitalisation, and digital public services (Fura et al., 2025; Privara & Caplanova, 2025). Rather than directly generating profitability, these macro-digital conditions shape how effectively cybersecurity governance practices are implemented, monitored, and enforced (Calliess & Baumgarten, 2020).

From a governance perspective, cybersecurity capabilities represent organisational control mechanisms that reduce operational disruption risk, strengthen regulatory compliance, and improve monitoring efficiency. CSD serves as a visible manifestation of these governance practices by increasing transparency around cyber-risk management systems, oversight structures, and incident-response capabilities. By strengthening managerial accountability and supervisory oversight, CSD can improve operational stability and cost efficiency, thereby contributing to financial performance through governance channels rather than reputational or signaling effects alone (Bruno et al., 2025; Calderon & Gao, 2020).

Despite these developments, empirical evidence linking CSD to bank profitability across different digital-institutional environments remains limited. Existing research rarely examines whether national digital maturity conditions the effectiveness of cybersecurity governance mechanisms (Amani et al., 2025). Digital maturity influences infrastructure quality, supervisory capacity, and digital human capital, all of which shape whether CSD reflects substantive governance implementation or symbolic reporting behaviour (Fura et al., 2025; Lafuente et al., 2024).

To address this gap, this study examines whether CSD is associated with bank profitability and whether national digital maturity conditions this relationship. Using a panel dataset of 889 banks across 21 European countries from 2014 to 2023, the study develops a dual disclosure measure capturing both the presence and intensity of cybersecurity reporting. A multi-method empirical strategy—combining fixed-effects estimation with complementary robustness and endogeneity checks—is applied to address unobserved heterogeneity, selection concerns, and potential reverse causality. The analysis also explores whether the conditioning role of digital maturity varies by bank size, reflecting differences in cyber exposure and governance complexity.

This study contributes to the literature in several ways. First, it shifts attention from market-based outcomes toward realised operational performance, providing direct evidence on the governance consequences of CSD. Second, it adopts an agency-based governance perspective in which CSD improves profitability through monitoring discipline, compliance efficiency, and disruption-risk reduction, while institutional digital maturity explains cross-country variation in governance effectiveness. Third, it introduces national digital maturity as a macro-institutional boundary condition shaping the operational impact of cybersecurity governance rather than acting as a direct performance driver. Collectively, these contributions provide new insights into how cybersecurity transparency supports operational resilience within the European banking system.

2. Literature review

2.1. Cybersecurity disclosure and bank performance

The digital transformation of banking has fundamentally reshaped how financial institutions deliver services, manage information flows, and control operational risk. The increasing integration of cloud computing, artificial intelligence, open-banking systems, and digital payment platforms has enhanced efficiency and service accessibility (Allahham et al., 2024a; Khalaf et al., 2023). At the same time, this digital integration has expanded banks' exposure to cybersecurity threats, including ransomware attacks, data breaches, phishing campaigns, and supply-chain intrusions that can disrupt service continuity and impose substantial compliance and recovery costs (Allahham et al., 2024a; Elsayed et al., 2024; ENISA, 2023; Tabash et al., 2025). Within this risk-intensive environment, CSD has emerged as a governance tool through which banks communicate how cyber risks are identified, monitored, and managed.

Early research often conceptualised CSD primarily as a transparency mechanism aimed at reducing stakeholder uncertainty and supporting organisational legitimacy. Transparent cybersecurity reporting may signal that banks recognise digital threats and maintain formal risk-management systems designed to protect customers and operational infrastructure (Hasani et al., 2023; Nelson & Wang, 2024). Empirical evidence suggests that cybersecurity transparency can strengthen customer confidence and institutional trust, particularly in rapidly digitalising financial systems (Cele & Kwenda, 2025; Tan et al., 2025). While these signaling and legitimacy effects are relevant, they offer limited insight into whether CSD generates measurable operational value within banking organisations.

From a governance perspective, agency theory provides a more direct explanation of how CSD may influence bank profitability. Cyber-risk management involves complex technical choices, significant investment commitments, and substantial managerial discretion, creating potential agency conflicts between managers, shareholders, and other stakeholders. Disclosure of cybersecurity governance arrangements—such as oversight structures, internal control systems, and incident-response frameworks—reduces information asymmetry and strengthens monitoring by boards, regulators, and supervisory authorities (Alodat et al., 2024; Barry et al., 2022; Boggini, 2024). Banks with stronger governance oversight, including technologically competent directors and audit committees with cybersecurity expertise, tend to provide more comprehensive cybersecurity reporting, which enhances accountability and improves internal risk monitoring (Guohong et al., 2025). Through these governance channels, CSD can reduce disruption-related costs, improve compliance efficiency, and enhance operational stability, thereby supporting profitability independently of market-based signaling effects.

Empirical evidence largely supports this governance-based interpretation of CSD, although findings remain context-dependent. Several studies report that more detailed cybersecurity reporting is associated with stronger risk-management practices, fewer operational disruptions, and closer alignment between digital risk controls and organisational sustainability objectives (Bruno et al., 2025; Tan et al., 2025). Conversely, other studies suggest that disclosure may increase regulatory scrutiny or unintentionally expose vulnerabilities, potentially weakening financial performance in certain institutional settings (Barry et al., 2022). These mixed results indicate that the economic consequences of CSD depend not only on disclosure practices themselves but also on the governance and institutional environments in which they are embedded.

Despite increasing scholarly interest in cybersecurity transparency, most prior research continues to emphasise market reactions and investor-based valuation outcomes rather than realised operational performance. This limitation is particularly salient in the banking sector, where cyber incidents directly affect service delivery, regulatory compliance costs, and organisational resilience. Profitability therefore represents a more appropriate outcome for evaluating whether cybersecurity governance mechanisms generate tangible financial benefits. Moreover, cross-country evidence on the relationship between CSD and bank profitability remains limited, especially within Europe's highly integrated and digitally advanced banking ecosystem.

Taken together, existing theory and evidence suggest that CSD functions primarily as a governance mechanism that enhances monitoring, strengthens internal controls, and mitigates cyber-related

operational risk. However, the extent to which these governance benefits translate into realised bank profitability remains insufficiently examined in a European context. Addressing this gap, the present study investigates whether is positively associated with bank profitability. Accordingly, the following hypothesis is proposed:

H1. Cybersecurity disclosure is positively associated with bank profitability.

2.2. Role of digital economy and society index (DESI)

While CSD reflects internal governance quality, its effectiveness depends on the broader digital environment in which banks operate. National digital maturity shapes the technological infrastructure, regulatory coordination, and supervisory capacity that determine whether cybersecurity governance practices can be effectively implemented, monitored, and enforced. Mixed empirical findings in the CSD literature suggest that institutional readiness and digital capability function as contextual boundary conditions influencing the effectiveness of governance mechanisms across countries (Bruno et al., 2025; Hasani et al., 2023; Masoud & Al-Utaibi, 2022; Tan et al., 2025). To capture this cross-country variation, this study introduces the DESI a national-level conditioning factor.

Although DESI is measured at the country level (Tsakalerou, 2025), its relevance for bank-level outcomes arises from the strong dependence of financial institutions on national digital infrastructures and regulatory ecosystems. Institutional features captured by DESI—including broadband connectivity, digital skills, cybersecurity preparedness, regulatory interoperability, and digital public services—shape the reliability, credibility, and enforceability of cybersecurity governance practices. In digitally mature environments, integrated infrastructure and coordinated regulatory frameworks facilitate real-time cyber monitoring, efficient information flows, and effective supervisory oversight. These conditions enhance the operational credibility of CSD by ensuring that reported governance practices are verifiable and consistently applied. By contrast, in digitally less mature environments, fragmented infrastructure and weaker regulatory coordination reduce the practical governance value of disclosure, increasing the likelihood that cybersecurity reporting remains symbolic rather than operational.

This moderating role of DESI is consistent with institutional contingency logic, which emphasises that firm-level governance mechanisms operate within, and are constrained by, broader regulatory and technological environments. Prior research demonstrates that national digital readiness conditions firm-level outcomes in areas such as innovation, ESG performance, digital transformation, and risk management by providing complementary institutional capacities that enable organisational effectiveness (Lafuente et al., 2024; Privara & Caplanova, 2025). In the banking sector, digital infrastructure quality and regulatory digitalisation influence whether cybersecurity governance practices can be effectively implemented, audited, and enforced across institutions.

Importantly, DESI is not conceptualised as a direct determinant of bank profitability. Rather, it captures institutional conditions that shape whether CSD reflects substantive governance implementation or merely formal compliance. Strong national digital environments reduce coordination costs, improve information transparency, and support regulatory enforcement, thereby enabling banks to translate cybersecurity governance practices into enhanced operational resilience and compliance efficiency (Buglea et al., 2025; Fura et al., 2025; Imran et al., 2022; Laitsou et al., 2025). Conversely, lower levels of digital maturity constrain banks' ability to operationalise cybersecurity governance even when disclosure is present.

Developed by the European Commission, DESI provides a comprehensive and policy-validated measure of digital readiness across five dimensions: connectivity, human capital, internet use, business digitalisation, and digital public services. Empirical studies confirm that DESI captures national digital capabilities associated with productivity, innovation diffusion, and digital governance quality across European economies (Afolabi, 2025; Laitsou et al., 2025; Privara & Caplanova, 2025; Vărzaru, 2024). These institutional digital capacities constitute external conditions that complement internal cybersecurity governance resources at the firm level.

From an institutional complementarity perspective, CSD can be viewed as an intangible governance capability whose effectiveness depends on alignment with national digital infrastructure and regulatory capacity. In digitally mature jurisdictions, high-quality infrastructure, skilled digital labour, and coordinated supervisory oversight enable banks to operationalise cybersecurity governance and derive measurable performance benefits. In less digitally mature contexts, institutional capacity constraints limit these governance returns (Kurnia & Ardianto, 2024; Wang et al., 2024). Accordingly, DESI is positioned as a boundary condition that shapes the consistency and credibility of cybersecurity governance rather than as a general driver of bank performance. Based on this framework, the study proposes the following hypothesis:

H2. The positive association between CSD and bank profitability is stronger in countries with higher levels of digital maturity.

2.3. Theoretical framework

Consistent with prior research integrating agency-based governance mechanisms with institutional conditioning (e.g., Barry et al., 2022; Imran et al., 2022), this study conceptualises CSD as a firm-level monitoring tool whose effectiveness depends on the surrounding digital–institutional environment. The relationship between CSD and bank profitability is therefore best understood through a governance-centred, multilevel framework that links internal monitoring and control mechanisms with national digital maturity. Within this framework, profitability reflects the extent to which governance practices reduce operational risk, enhance efficiency, and support regulatory compliance, rather than short-term market perceptions (Amani et al., 2025; Abdurrahman, 2024).

A central distinction underpinning this framework is between market valuation responses and realised profitability outcomes. While prior studies show that cybersecurity announcements and disclosures may influence investor perceptions and short-term stock market reactions (Berkman et al., 2018), such effects are often transitory and sentiment-driven. In contrast, accounting-based profitability captures more persistent outcomes associated with internal efficiency, risk-management quality, supervisory discipline, and governance effectiveness (Bruno et al., 2025). Accordingly, this study focuses on profitability as the primary channel through which the economic consequences of cybersecurity governance materialise.

From an agency-theoretic perspective, cybersecurity risks and internal control weaknesses are inherently opaque, creating information asymmetries that elevate agency costs. In the absence of effective monitoring, managers may underinvest in cybersecurity capabilities or obscure deficiencies in cyber governance, particularly given the technical complexity and limited observability of cyber risks. CSD mitigates these agency problems by increasing transparency around governance structures, oversight arrangements, and risk-management processes related to cyber threats (Calderon & Gao, 2020). Enhanced disclosure strengthens monitoring by boards, regulators, and supervisory authorities, constrains managerial opportunism, and reinforces internal accountability. Through these mechanisms, CSD contributes to improved governance discipline, greater operational resilience, and ultimately higher and more stable profitability (Bruno et al., 2025).

The effectiveness of these governance mechanisms, however, is not uniform across institutional environments. Institutional theory emphasises that firm-level practices are embedded within national contexts that shape how governance mechanisms are implemented, monitored, and enforced. Digital transformation does not generate homogeneous outcomes across countries because institutional conditions—such as digital infrastructure quality, human capital, supervisory capacity,

regulatory coherence, and enforcement credibility—vary substantially (Imran et al., 2022; Laitsou et al., 2025). These conditions determine whether CSD operates as a substantive governance mechanism or remains largely symbolic.

Within this framework, national digital maturity—captured by the DESI—represents a set of institutional boundary conditions that condition the effectiveness of cybersecurity governance. Digital maturity is not conceptualised as a firm-level capability and is not expected to directly generate bank profitability. Rather, it shapes the institutional environment in which cybersecurity governance practices are verified, enforced, and operationalised. In digitally mature environments, stronger infrastructure, higher levels of digital expertise, and more coordinated supervisory frameworks enhance the credibility and operational relevance of CSD. Under such conditions, governance improvements associated with disclosure are more likely to translate into tangible efficiency gains, reduced operational disruptions, and improved risk management, thereby strengthening profitability (Privara & Caplanova, 2025). Conversely, in environments characterised by weaker digital infrastructure and limited supervisory capacity, the governance benefits of disclosure are attenuated, reducing their impact on realised performance.

Taken together, this framework positions CSD as a firm-level governance mechanism whose profitability effects arise primarily through improved monitoring, reduced agency costs, and enhanced operational resilience (Mansour et al., 2026). Institutional theory explains why these effects vary across countries, as national digital maturity conditions the extent to which cybersecurity governance practices are credible, enforceable, and economically consequential. Fig. 1 illustrates this framework, depicting digital maturity (DESI) as a moderating institutional condition that strengthens or weakens the relationship between CSD and bank profitability.

3. Method

3.1. Sample and data description

This study employs a multi-country panel design to examine how CSD affects bank financial performance, conditional on national digital maturity measured by the European Commission's DESI. We initially extracted data for 1413 listed and large banks from 25 European countries for the period 2014–2023 from Refinitiv Eikon (Thomson Reuters) (https://www.refinitiv.com). After excluding banks with missing financial, ESG, or governance information and countries without full DESI coverage for the study period, we obtained a balanced panel of 889 banks from 21 EU countries, yielding 8890 bank-year observations.

Table 1 reports the country distribution. The sample covers major banking systems (Germany, France, the Netherlands, Sweden, Ireland) and smaller emerging EU markets (Romania, Croatia, Estonia), allowing us to test the CSD–performance relationship across heterogeneous institutional contexts (Sun et al., 2023).

Bank-level financial and governance data (ROA, ROE, total assets, age, leverage, board attributes, and audit-committee characteristics) were collected from Refinitiv Eikon. CSD indicators were obtained from the same source and cross-checked with banks' annual, sustainability, or regulatory reports when necessary. DESI data were retrieved from the European Commission's Digital Strategy Portal (https://digital-strategy.ec.europa.eu/en/policies/desi), which reports annual scores for connectivity, human capital, internet use, business digitalization, and digital public services.

Country-level controls — GDP growth, inflation, and Worldwide Governance Indicators (WGI) — were obtained from the World Bank Open Data platform (https://data.worldbank.org).

Europe's banking sector has, over the past decade, been exposed to high-impact cyber incidents and to new regulatory requirements. Incidents involving major European and UK institutions (e.g. the 2016 Tesco Bank breach, ECB website compromise, attacks on Dutch banks, and the 2023 MOVEit supply-chain incident) accelerated the EU's regulatory responses, particularly NIS2 (2023). These frameworks, together with guidance from ENISA, the ECB, and the EBA, have made CSD a core element of prudential and governance practice, which justifies focusing on the 2014–2023 window.

3.2. Variable measurement, description, and definitions

This study operationalizes constructs at both the firm and national

Table 1
Distribution of European banks.

	Country	No. of Bank	Obs.		Country	No. of Bank	Obs.
1	Spain	32	320	12	Portugal	34	340
2	Belgium	24	240	13	Poland	26	260
3	Germany	118	1180	14	Romania	22	220
4	Italy	24	240	15	Greece	40	400
5	Austria	24	240	16	Czech Republic	38	380
6	Netherlands	52	520	17	Slovenia	28	280
7	France	62	620	18	Luxembourg	38	380
8	Ireland	66	660	19	Hungary	42	420
9	Denmark	44	440	20	Croatia	22	220
10	Sweden	84	840	21	Estonia	29	290
11	Finland	40	400		**Total**	**889**	**8890**

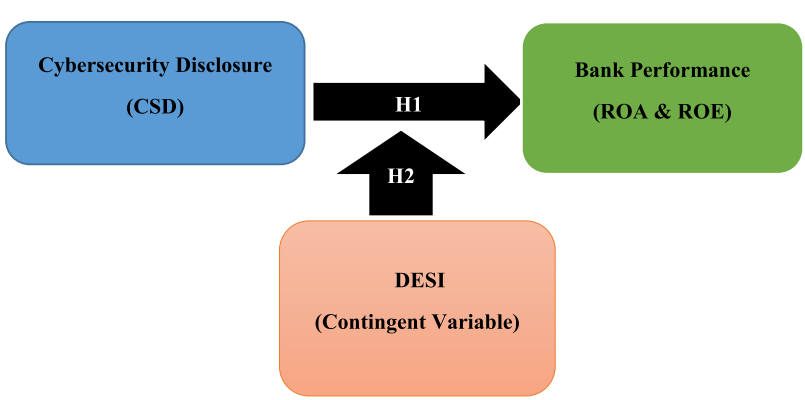

Fig. 1. Theoretical framework: moderator role of digital maturity (DESI).
This framework positions CSD as a firm-level governance mechanism whose direct effect on bank profitability is primarily explained by agency-based monitoring and control considerations, while Institutional Theory—operationalized through the DES—captures the macro-digital conditions that condition the strength of this relationship.

levels to ensure theoretical and measurement consistency between cybersecurity governance and financial performance. Table 2 summarizes all variables, definitions, and data sources.

3.2.1. Dependent variables

Bank performance is evaluated using ROA and ROE — two accounting-based profitability indicators widely used in banking and corporate-governance research (Bruno et al., 2025; Saleh et al., 2026; Ziaei & Roshani, 2024). ROA captures the internal efficiency with which banks transform assets into earnings; ROE reflects the external profitability delivered to shareholders and is consistent with signaling arguments, where credible disclosure strengthens investor confidence. Unlike market-based indicators such as Tobin's Q, ROA and ROE represent realized operating outcomes and are therefore aligned with this study's focus on the performance effects of governance-related disclosure (Alhasnawi et al., 2025; Elsayed et al., 2024). Collectively, both indicators offer a clear, comprehensive, and empirically stable assessment of bank performance across countries and over time (Mansour et al., 2025).

3.2.2. Independent variable

CSD has emerged as an important governance mechanism through which banks communicate their exposure to digital risks and the organisational structures established to manage them. Building on prior research on disclosure and cybersecurity governance (Alodat et al., 2024; Mansour et al., 2026a; Radu & Smaili, 2022), this study operationalises CSD using two complementary indicators designed to capture both the existence and the intensity of cybersecurity-related reporting.

The first indicator is a binary measure reflecting the presence of CSD. A bank is assigned a value of one if it discloses cybersecurity-related information—such as cyber-risk policies, risk-management systems, incident reporting, or board-level oversight—in its annual, sustainability, or regulatory reports, and zero otherwise. This measure provides a consistent baseline for identifying whether banks engage in formal cybersecurity transparency across European jurisdictions characterised by heterogeneous disclosure requirements.

To capture variation in disclosure intensity, a second continuous indicator (CSD_Volume) measures the extent of cybersecurity-related reporting. Following Smaili et al. (2023), this variable reflects the number of words or paragraphs in each annual report containing cybersecurity-related keywords (e.g., *cyber, cybersecurity, information security, data protection*). By capturing the prominence and persistence of cybersecurity reporting, this measure distinguishes between minimal references and more sustained engagement with cyber-risk governance.

We acknowledge that volume-based measures capture disclosure extent rather than qualitative depth and therefore cannot fully differentiate between informative and boilerplate reporting. However, in the context of cybersecurity governance—where disclosure remains largely voluntary, technically complex, and weakly standardised—prior research interprets disclosure volume as a reasonable proxy for **managerial attention, transparency commitment**, and the **organisational salience** of cyber risk (Radu & Smaili, 2022; Smaili et al., 2023). Consistent with agency-based governance logic, repeated and structured discussion of cybersecurity governance, internal controls, and oversight mechanisms is more likely to reflect embedded monitoring and accountability practices than isolated or generic statements (Elsayed et al., 2024; Gao et al., 2020).

To illustrate, low-intensity disclosure typically consists of brief references to cybersecurity risks within general operational risk sections, whereas high-intensity disclosure includes dedicated discussions of cyber-risk management frameworks, governance oversight arrangements, cybersecurity investments, incident-response procedures, and alignment with regulatory initiatives such as the DORA (Calderon & Gao, 2021). The volume-based indicator captures these differences by measuring how prominently cybersecurity topics are addressed within banks' corporate reporting.

All reports were independently coded by two trained researchers, and inter-coder reliability exceeded 95 percent, indicating a high level of consistency in content classification (Alodat et al., 2024). Taken

Table 2
Variables description & definitions.

Variable Type	Variable Name	Label / Code	Description and Measurement	Source
Dependent Variables	Return on Assets	ROA $_{i,t}$	Net income divided by total assets for bank i in period t.	Refinitiv Eikon
	Return on Equity	ROE $_{i,t}$	Net income divided by shareholders' equity for bank i in period t.	Refinitiv Eikon
Independent Variables	Cybersecurity Disclosure (Binary)	CSD_Binary $_{i,t}$	Dummy = 1 if bank i discloses any cybersecurity-related information (policy, risk management, incidents, board oversight) in reports during period t; 0 otherwise (Radu & Smaili, 2022).	Refinitiv Eikon / Annual Reports
	Cybersecurity Disclosure (Volume)	CSD_Volume$_{i,t}$	Continuous measure of the volume of cybersecurity-related text in bank i's annual report for year t, defined as the total number of words or paragraphs containing cybersecurity-related keywords ("cyber," "cybersecurity," "information security," etc.). Higher values indicate more intensive and transparent disclosure (Smaili et al., 2023).	Refinitiv Eikon / Manual Coding
Moderating Variable	Digital Economy and Society Index	DESI $_t$	A composite index published by the European Commission measuring national digital maturity (0–100) across five pillars: connectivity, human capital, internet use, business digitalization, and digital public services.	European Commission
Control Variables	Audit Committee Expertise	AC_Expertise $_{i,t}$	Equals 1 if the audit committee has at least three members including one financial expert, and 0 otherwise; captures audit oversight quality.	Refinitiv Eikon
	Board Gender Diversity	BGD $_{i,t}$	Percentage of female directors on the bank's board; reflects gender diversity and inclusivity in governance.	Refinitiv Eikon
	Board Independence	B_Independent $_{i,t}$	Percentage of independent directors serving on the bank's board, representing external oversight strength.	Refinitiv Eikon
	Board Skills	B_Skills $_{i,t}$	Percentage of board members with expertise in finance, risk management, or corporate governance.	Refinitiv Eikon
	Bank Size	B_SIZE $_{i,t}$	Natural logarithm of total bank assets; controls for firm size effects.	Refinitiv Eikon
	Bank Age	B_AGE $_{i,t}$	Natural logarithm of the number of years since the bank's establishment; proxies organizational maturity.	Refinitiv Eikon
	Bank Leverage	B_LEV $_{i,t}$	Ratio of total debt to total equity; reflects financial risk.	Refinitiv Eikon
	Inflation	INF $_t$	Annual inflation rate (%) of the bank's home country; macroeconomic control.	World Bank
	GDP Growth	GDP $_t$	Annual real GDP growth rate (%) of the bank's home country; proxies economic environment.	World Bank
	Worldwide Governance Indicators	WGI $_t$	Composite governance index (–2.5 to +2.5) from the World Bank measuring institutional quality, including regulatory quality, rule of law, corruption control, and government effectiveness.	World Bank

together, the dual-measure approach captures both the presence and relative intensity of CSD, providing a nuanced representation of cybersecurity transparency that is well suited to empirical analysis while remaining consistent with established disclosure research practices.

3.2.3. Moderating variable

National digital maturity is measured using the DESI, which captures the institutional digital environment in which banks operate (Vărzaru, 2024). Although DESI is constructed at the country level, it is relevant for bank-level outcomes because financial institutions rely on shared digital infrastructure, regulatory coordination, supervisory capacity, and knowledge diffusion mechanisms that shape how cybersecurity governance practices are implemented, monitored, and enforced across jurisdictions (Mansour et al., 2026; Privara & Caplanova, 2025a). In the banking sector, national digital maturity influences the reliability of cybersecurity infrastructure, the integration of digital public services, and the effectiveness of data-governance regimes supporting secure financial operations and regulatory compliance (Afolabi, 2025; Bianchini et al., 2023).

Developed by the European Commission, DESI is a composite index ranging from 0 to 100 and comprising five pillars: connectivity, human capital, internet use, business digitalisation, and digital public services (Mansour et al., 2026a). It provides a comprehensive and policy-validated proxy for national digital readiness and institutional digital capacity within the European Union (Imran et al., 2022). Consistent with its conceptual design, DESI is employed in this study as a contextual moderating variable, reflecting external digital–institutional conditions rather than firm-level technological capability or managerial strategy. Prior empirical research shows that DESI explains cross-country variation in innovation diffusion, regulatory effectiveness, and digital governance quality across EU member states (Laitsou et al., 2025; Vărzaru, 2024).

From an institutional perspective, national digital maturity is not expected to directly generate bank profitability. Rather, it represents a set of institutional boundary conditions that shape the credibility, consistency, and enforceability of firm-level cybersecurity governance practices. Digitally mature environments typically provide more robust supervisory frameworks, interoperable digital infrastructure, and higher levels of digital expertise, which enhance the operational relevance of CSD and support more effective implementation of risk-management systems (Constantin et al., 2023; Fura et al., 2025; Privara & Caplanova, 2025). Conversely, lower levels of digital maturity weaken oversight coordination and infrastructure reliability, reducing the practical governance value of disclosure and increasing the likelihood that cybersecurity reporting remains largely symbolic.

Importantly, DESI does not capture bank-specific governance quality, managerial capability, or competitive strategy, nor does it explain performance differences among banks operating within the same country. Accordingly, this study interprets DESI strictly as shaping the institutional environment in which CSD operates, rather than as a general enhancer of bank profitability. This interpretation is consistent with the study's multilevel theoretical framework and supports the use of DESI as a moderating, rather than explanatory, performance variable.

3.2.4. Control variables

Control variables account for governance, firm, and macroeconomic heterogeneity influencing financial performance (Cortez & Dekker, 2022):

Governance factors—audit-committee expertise, board gender diversity, independence, and skills capture monitoring quality and decision diversity (Al-Tahat et al., 2025; Elsayed et al., 2024; Firoozi & Mohsni, 2023).

- Firm attributes—size, age, and leverage control for scale efficiency, organizational maturity, and financial risk (Berkman et al., 2018; Smaili et al., 2023).
- Macroeconomic indicators—GDP growth, inflation, and WGI reflect cyclical conditions and institutional quality (Gao et al., 2020; Héroux & Fortin, 2024; World Bank, 2023).

Year- and firm-fixed effects absorb unobserved heterogeneity and time shocks (Saleh & Mansour, 2024). Firm-level data come from Refinitiv Eikon, and macro-data from the World Bank (2023).

3.3. Statistical model

This study estimates the direct effect of CSD on bank profitability and examines whether this relationship is conditioned by national digital maturity. Variable definitions are provided in Table 2.

Model 1: Direct Effect of CSD

$$PERFORMANCE_{i,t} = \beta_0 + \beta_1 CSD_{i,t} + \beta_2 DESI_t + \beta_3 AC.Expertise_{i,t} + \beta_4 BGD_{i,t} + \beta_5 B.Independent_{i,t} + \beta_6 Bskills_{i,t} + \beta_7 B.SIZE_{i,t} + \beta_8 B.AGE_{i,t} + \beta_9 B.LEV_{i,t} + \beta_{10} INF_t + \beta_{11} GDP_t + \beta_{12} WGI_t + \epsilon_{i,t} \quad (1)$$

Model 2: Moderating Effect of DESI

$$PERFORMANCE_{i,t} = \beta_0 + \beta_1 CSD_{i,t} + \beta_2 DESI_t + \beta_3 CSD \times DESI_t + \beta_4 AC_Expertise_{i,t} + \beta_5 BGD_{i,t} + \beta_6 B_Independent_{i,t} + \beta_7 B_skills_{i,t} + \beta_8 B_SIZE_{i,t} + \beta_9 B_AGE_{i,t} + \beta_{10} \beta_LEV_{i,t} + \beta_{11} INF_t + \beta_{12} GDP_t + \beta_{13} WGI_t + \epsilon_{i,t} \quad (2)$$

Where PERFORMANCE denotes bank financial performance, measured alternately by ROA and ROE. β0 is the intercept and $\epsilon_{i,t}$ is the error term. The CSD variables include both CSD_Binary and CSD_Volume (word/paragraph-based disclosure intensity). DESI, a country-level construct, represents macro-digital readiness and is merged annually with firm-level data. The interaction term (CSD*DESI) tests H2 that disclosure yields higher profitability in digitally mature economies.

All baseline models are estimated using bank- and year-fixed effects to control for unobserved, time-invariant heterogeneity and common macroeconomic shocks (Al-Mohareb, 2025; Smaili et al., 2023). However, fixed-effects estimation alone may be insufficient in this context due to potential reverse causality between performance and disclosure, the persistence of governance practices, and non-random disclosure decisions.

To address these concerns, the analysis adopts a **sequential, problem-driven estimation strategy**. First, a Heckman two-step selection model (Heckman, 1979) corrects for disclosure-selection bias by accounting for the non-random likelihood that banks choose to disclose cybersecurity information, consistent with evidence that cybersecurity reporting is strategically driven (Masoud & Al-Utaibi, 2022). Second, two-step System GMM (Arellano & Bond, 1991; Blundell & Bond, 1998) is employed to address dynamic endogeneity and feedback effects arising from the persistence of profitability and governance practices (Kurnia & Ardianto, 2024; Saleh et al., 2026). Third, propensity score matching (PSM; Rosenbaum & Rubin, 1985) compares high-CSD banks with observationally similar low-CSD banks based on governance

Table 3
Descriptive statistics.

Variables	Mean	Std. Dev.	Min	Max	Skew.	Kurt.
ROA	0.026	0.032	0.002	0.124	1.885	5.933
ROE	0.132	0.084	0.029	0.344	1.033	3.369
CSD_{Binary}	0.599	0.473	0.000	1.000	−0.520	1.313
DESI	49.915	9.374	31.189	69.202	−0.581	0.460
AC_Expertise	0.902	0.352	0.000	0.987	−2.106	5.302
BGD	32.836	10.705	14.783	52.002	−0.128	2.065
B_Independent	68.617	19.071	31.107	103.630	−0.070	2.372
B_Skills	42.095	20.570	10.028	72.959	0.030	1.949
B_SIZE	25.192	2.260	19.550	27.300	−0.157	2.153
B_AGE	35.287	35.396	2.050	172.329	1.794	6.775
B_LEV	1.393	2.012	0.065	4.868	1.084	2.334
INF	2.468	1.125	0.220	5.610	0.533	2.229
GDP	1.942	1.947	−4.255	5.836	−0.436	2.782
WGI	1.324	0.693	−0.225	2.420	−0.357	2.573

characteristics, size, leverage, and macroeconomic conditions, thereby mitigating observable selection bias (Li et al., 2018).

Each estimation technique targets a distinct source of bias—unobserved heterogeneity, dynamic endogeneity, and non-random disclosure behavior. Consistency of results across these complementary approaches strengthens causal interpretation while avoiding reliance on any single estimator. All estimations are conducted using STATA 14.

4. Results

4.1. Descriptive statistics

Table 3 reports descriptive statistics for 8890 bank-year observations. European banks exhibit stable profitability, with mean ROA of 2.6 % and ROE of 13.2 %, although positive skewness indicates performance heterogeneity across institutions. Approximately 60 % of banks disclose cybersecurity information, suggesting widespread but uneven adoption of cyber governance practices. DESI values indicate moderate but diverse national digital maturity, providing suitable cross-country variation for testing institutional moderation effects. Governance indicators reflect generally strong supervisory structures, although board skill dispersion suggests differences in governance capacity. Overall, the data show sufficient variation across disclosure, governance, and institutional digital environments to support multivariate analysis.

4.2. Bivariate correlations

Table 4 presents pairwise correlations among the variables. CSD is positively associated with profitability, indicating that banks with greater cyber transparency tend to exhibit stronger performance. National digital maturity is also positively related to disclosure and profitability, suggesting that digitally developed institutional environments support cybersecurity governance and financial outcomes. Governance variables generally correlate positively with performance, while leverage shows a negative association. Correlation levels remain below multicollinearity thresholds, supporting the reliability of subsequent regression analysis (Tabash et al., 2024).

4.3. Multicollinearity assessment

Variance Inflation Factors (VIF) were computed to check for possible multicollinearity among the independent variables. All VIF values were well under 5, indicating that multicollinearity does not compromise the stability of the estimates. In summary, these diagnostics verify that multicollinearity is not an issue, allowing us to keep all explanatory variables in the regression model (Table 5).

Table 4
Pairwise correlations.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
(1) ROA	1.000													
(2) ROE	0.100* (0.001)	1.000												
(3) CSD_{Binary}	0.324* (0.001)	0.411* (0.001)	1.000											
(4) DESI	0.093* (0.001)	0.055* (0.009)	0.284* (0.001)	1.000										
(5) AC_Expertise	0.153* (0.001)	0.262* (0.001)	0.035 (0.371)	0.131* (0.001)	1.000									
(6) BGD	0.177* (0.001)	0.133* (0.001)	−0.021 (0.616)	0.042* (0.028)	0.122* (0.003)	1.000								
(7) B_Independent	−0.016 (0.714)	0.131* (0.002)	0.003 (0.941)	0.182* (0.001)	0.041 (0.284)	0.103* (0.011)	1.000							
(8) B_Skills	0.232* (0.001)	0.146* (0.001)	−0.007 (0.853)	0.115* (0.001)	0.370* (0.001)	−0.062 (0.106)	0.037 (0.381)	1.000						
(9) B_SIZE	0.269* (0.001)	0.199* (0.001)	0.265* (0.001)	−0.057* (0.002)	0.048 (0.234)	0.211* (0.001)	0.204* (0.001)	−0.017 (0.660)	1.000					
(10) B_AGE	0.217* (0.001)	0.082* (0.039)	0.042 (0.293)	0.170* (0.001)	−0.035 (0.365)	0.094* (0.027)	−0.124* (0.001)	−0.030 (0.415)	0.241* (0.001)	1.000				
(11) B_LEV	−0.058* (0.003)	−0.054* (0.001)	0.536* (0.001)	−0.195* (0.001)	0.112* (0.001)	0.112* (0.001)	0.046* (0.021)	0.041* (0.037)	0.111* (0.001)	−0.048* (0.018)	1.000			
(12) INF	−0.043 (0.241)	−0.035 (0.362)	−0.017 (0.489)	−0.261* (0.001)	0.022 (0.641)	−0.026 (0.581)	0.044 (0.254)	0.030 (0.417)	−0.052 (0.132)	0.023 (0.642)	0.031 (0.401)	1.000		
(13) GDP	0.041 (0.263)	0.030 (0.416)	0.018 (0.486)	0.295* (0.001)	0.006 (0.887)	0.017 (0.481)	0.052 (0.138)	0.020 (0.471)	0.034 (0.390)	0.031 (0.401)	−0.038 (0.352)	−0.387* (0.001)	1.000	
(14) WGI	0.113* (0.004)	0.105* (0.006)	0.064* (0.014)	0.402* (0.001)	0.048 (0.203)	0.076* (0.009)	0.089* (0.003)	0.058 (0.113)	0.052 (0.132)	0.092* (0.019)	−0.071 (0.091)	−0.284* (0.001)	0.321* (0.001)	1.000

Note: * $P < 0.05$ (2-tailed).

Table 5
VIF analysis.

Variable	VIF	
	$CSD_{_Binary}$	$CSD_{_Volume}$
$CSD_{_Binary}$	1.05	1.08
DESI	1.10	1.12
AC_Expertise	1.27	1.33
BGD	1.14	1.16
B_Independent	1.17	1.19
B_Skills	1.31	1.34
B_SIZE	1.26	1.29
B_AGE	1.09	1.10
B_LEV	1.13	1.15
GDP	1.08	1.09
INF	1.04	1.05
WGI	1.09	1.11
Mean VIF	**1.21**	**1.24**

Table 6
Results of panel regression analysis.

VARIABLES	Model 1		Model 2	
	ROA	ROE	ROA	ROE
Constant	3.438 ***	3.499 ***	0.210 ***	2.398 ***
	(0.470)	(0.466)	(0.013)	(0.662)
$CSD_{_Binary}$	0.230 ***	0.342 ***	0.316 ***	0.293 ***
	(0.042)	(0.065)	(0.055)	(0.027)
DESI	0.268 ***	0.352 **	0.408 ***	0.345 ***
	(0.094)	(0.146)	(0.092)	(0.020)
$CSD_{_Binary}$*DESI	–	–	0.445 ***	0.362 ***
			(0.098)	(0.053)
AC_Expertise	0.215 ***	0.466 ***	0.250 ***	0.351 ***
	(0.042)	(0.093)	(0.034)	(0.020)
BGD	0.051 ***	0.100 ***	0.284 ***	0.347 ***
	(0.012)	(0.013)	(0.073)	(0.020)
B_Independent	0.038 ***	0.039 ***	0.269 ***	0.242 ***
	(0.006)	(0.005)	(0.059)	(0.080)
B_Skills	0.100 *	0.103 ***	0.213 ***	0.186 ***
	(0.056)	(0.014)	(0.029)	(0.033)
B_SIZE	–0.099***	–0.081***	–0.091***	–0.095***
	(0.011)	(0.009)	(0.010)	(0.031)
B_AGE	0.228 ***	0.343 ***	0.455 *	0.186 *
	(0.088)	(0.021)	(0.242)	(0.101)
B_LEV	–0.168***	–0.218***	–0.070***	–0.113***
	(0.022)	(0.029)	(0.021)	(0.021)
GDP	0.041 **	0.058 **	0.049 **	0.066 **
	(0.017)	(0.024)	(0.019)	(0.025)
INF	–0.033 *	–0.041 *	–0.037 *	–0.048 *
	(0.019)	(0.022)	(0.021)	(0.024)
WGI	0.117 **	0.132 **	0.126 **	0.140 **
	(0.057)	(0.061)	(0.059)	(0.065)
Breusch & Pagan Test	126.37 ***	138.52 ***	132.41 ***	142.63 ***
Hausman Test	97.54 ***	48.36 ***	76.11 ***	39.57 ***
F-test	261.43	356.90	268.51	348.74
Prob > *F*	0.000	0.000	0.000	0.000
R^2 (%)	26.3	28.4	47.6	43.1
No. of Groups	889	889	889	889
No. of Obs.	8 890	8 890	8 890	8 890

Note: Standard errors are in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All Models include year and country fixed effects.

4.4. Panel regression analysis

Table 6 reports fixed-effects regression results examining the relationship between CSD and bank profitability, as well as the moderating role of national digital maturity. The fixed-effects specification is preferred based on Hausman tests and controls for unobserved bank-level heterogeneity, such as organisational culture and internal governance practices. Diagnostic tests confirm model adequacy and strong joint explanatory power (Chen, 2020).

Model 1 indicates that banks engaging in CSD consistently exhibit higher profitability. Rather than reflecting symbolic reporting, this pattern suggests that cybersecurity transparency functions as an internal governance mechanism that strengthens monitoring effectiveness, enhances risk-management discipline, and improves operational resilience. This interpretation aligns with governance-based disclosure research showing that cybersecurity reporting reduces information asymmetry and strengthens oversight structures (Calderon & Gao, 2020; Smaili et al., 2023).

National digital maturity enters positively in the baseline specification; however, this effect is not interpreted as a direct performance driver (Mansour et al., 2026a). Instead, it reflects institutional digital capacity—such as infrastructure quality, supervisory coordination, and regulatory integration—that shapes the environment in which governance mechanisms operate (Imran et al., 2022).

Model 2 introduces the interaction between CSD and digital maturity. The positive interaction indicates that the profitability benefits associated with disclosure are significantly stronger in digitally mature institutional environments, supporting Hypothesis 2. This pattern suggests that advanced digital ecosystems enhance the credibility and operational effectiveness of cybersecurity governance practices, consistent with institutional complementarity arguments (Imran et al., 2022; Laitsou et al., 2025).

The increase in explanatory power following inclusion of the interaction term highlights the importance of cross-country institutional heterogeneity. In digitally advanced environments, CSD appears to reflect embedded governance systems capable of improving internal controls and risk mitigation. Conversely, in less digitally mature contexts, disclosure practices appear to carry weaker operational implications, limiting their ability to generate measurable profitability improvements.

Governance variables display consistent performance-enhancing effects across specifications. Audit committee expertise, board gender diversity, board independence, and board skill composition are positively associated with profitability, reinforcing agency-based arguments that effective oversight improves monitoring capacity and decision quality in managing cyber-related risks (Alodat et al., 2024; Bruno et al., 2025). In contrast, bank size and leverage exhibit negative associations with performance, suggesting that organisational complexity and financial risk exposure may reduce operational efficiency.

At the macroeconomic level, economic growth is positively associated with profitability, whereas inflation exerts a modest negative influence. Institutional quality also contributes positively to performance, highlighting the role of broader governance environments in supporting financial stability.

Fig. 2 illustrates the interaction between CSD and national digital maturity using simple slope analysis (Aiken et al., 1991). The plots demonstrate that the performance differential between disclosing and non-disclosing banks increases substantially as digital maturity rises. In digitally advanced environments, disclosure appears to translate more effectively into performance gains, whereas in lower digital maturity settings the performance gap is comparatively limited. These visual results reinforce the interpretation that national digital maturity conditions the effectiveness of CSD as a governance mechanism.

4.5. Robustness check

To evaluate the stability and generalizability of the baseline findings, two complementary robustness tests are conducted. First, Earnings per Share (EPS) is used as an alternative performance indicator to verify that results are not driven by a specific profitability metric. Second, a continuous content-based disclosure measure (CSD_Volume) is introduced to examine whether disclosure depth, rather than disclosure presence alone, drives the observed relationships.

4.5.1. Alternative performance measure (EPS)

Table 7 presents regression results using EPS as an alternative

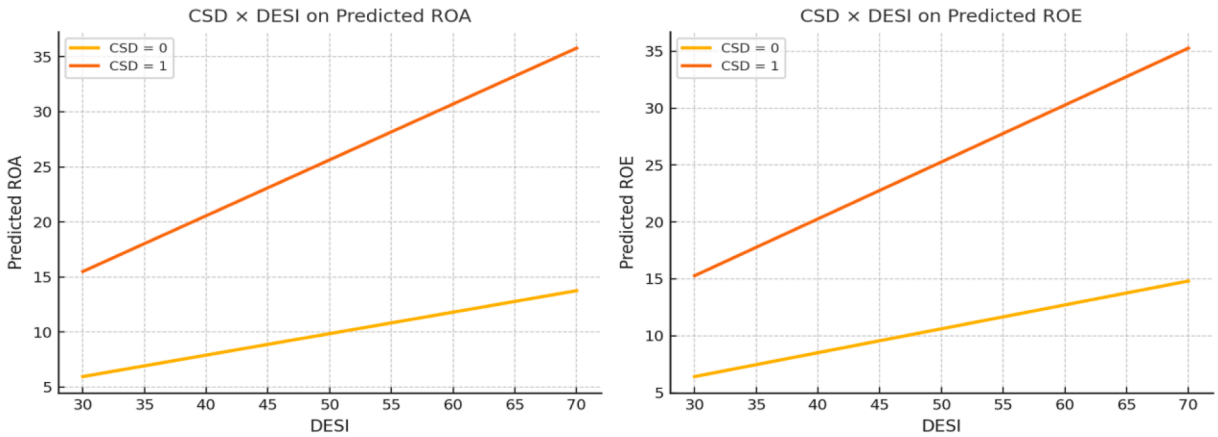

Fig. 2. CSD × DESI interaction effects on ROA and ROE.
Interaction effects of CSD and DESI on bank profitability (ROA and ROE).

Table 7
EPS regression analysis: alternative performance measure.

VARIABLES	Model 1 EPS	Model 2
Constant	0.076*** (0.027)	0.088*** (0.030)
$CSD_{_Binary}$	0.293*** (0.027)	0.316*** (0.055)
DESI	0.345*** (0.020)	0.408*** (0.092)
INTERACTION ($CSD_{_Binary}$*DESI)	- -	0.445*** (0.098)
BGD	0.347*** (0.020)	0.284*** (0.073)
B_LEV	−0.113*** (0.021)	−0.070*** (0.021)
B_SIZE	−0.095*** (0.031)	−0.091*** (0.010)
B_Independent	0.242*** (0.080)	0.269*** (0.059)
B_AGE	0.186* (0.101)	0.455* (0.242)
AC_Expertise	0.351*** (0.020)	0.250*** (0.034)
B_Skills	0.186*** (0.033)	0.213*** (0.029)
F-test	32.035	37.89
Prob > *F*	0.000	0.000
R^2	39.78 %	43.425 %
No. of Groups	889	889
Number of Observations	8890	8890

Note: Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. EPS is used as an alternative performance measure. Models include year and country fixed effects.

Table 8
Panel regression results.

VARIABLES	Model 1		Model 2	
	ROA	ROE	ROA	ROE
Constant	3.212 *** (0.458)	3.285 *** (0.470)	0.187 *** (0.015)	2.155 *** (0.620)
$CSD_{_Volume}$	0.188 *** (0.041)	0.276 *** (0.062)	0.251 *** (0.049)	0.239 *** (0.029)
DESI	0.243 ** (0.099)	0.321 ** (0.138)	0.392 *** (0.090)	0.327 *** (0.024)
$CSD_{_Volume}$*DESI	–	–	0.419 *** (0.090)	0.341 *** (0.049)
AC_Expertise	0.228 *** (0.040)	0.459 *** (0.090)	0.239 *** (0.031)	0.345 *** (0.018)
BGD	0.049 *** (0.011)	0.093 *** (0.012)	0.272 *** (0.070)	0.338 *** (0.019)
B_Independent	0.037 *** (0.006)	0.038 *** (0.005)	0.251 *** (0.057)	0.236 *** (0.075)
B_Skills	0.098 * (0.054)	0.099 *** (0.013)	0.207 *** (0.028)	0.179 *** (0.031)
B_SIZE	–0.097*** (0.010)	–0.080*** (0.008)	–0.089*** (0.009)	–0.092*** (0.029)
B_AGE	0.216 *** (0.082)	0.331 *** (0.020)	0.441 * (0.231)	0.174 * (0.098)
B_LEV	–0.161*** (0.021)	–0.205*** (0.028)	–0.065*** (0.020)	–0.107*** (0.019)
GDP	0.039 ** (0.016)	0.054 ** (0.022)	0.047 ** (0.018)	0.061 ** (0.024)
INF	–0.031 * (0.018)	–0.039 * (0.021)	–0.035 * (0.020)	–0.045 * (0.022)
WGI	0.112 ** (0.055)	0.127 ** (0.058)	0.119 ** (0.057)	0.135 ** (0.062)
Breusch & Pagan Test	124.96 ***	136.08 ***	130.87 ***	141.22 ***
Hausman Test	95.27 ***	46.28 ***	73.82 ***	37.40 ***
F-test	265.78	353.25	270.03	349.16
Prob > *F*	0.000	0.000	0.000	0.000
R^2 (%)	26.9	29.1	48.2	44.0
No. of Groups	889	889	889	889
No. of Obs.	8 890	8 890	8 890	8 890

Note: Standard errors are in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All Models include year and country fixed effects.

indicator of bank performance. The positive and statistically significant coefficient of CSD confirms that banks engaging in CSD consistently exhibit stronger earnings performance. This finding demonstrates that the relationship between cybersecurity transparency and financial outcomes extends beyond accounting efficiency measures such as ROA and ROE and is also reflected in shareholder-oriented earnings performance.

Importantly, the moderating effect of national digital maturity remains stable when EPS is used as the dependent variable. The positive interaction between CSD and DESI indicates that earnings benefits associated with disclosure are amplified in digitally mature institutional environments. This consistency strengthens the interpretation that digital infrastructure, supervisory coordination, and regulatory digital capacity enhance the operational effectiveness of cybersecurity governance mechanisms.

Governance and control variables remain stable across specifications, with oversight-related board attributes positively associated with earnings performance and bank size and leverage negatively related, reflecting organisational complexity and financial risk. The EPS results confirm that the effects of cybersecurity disclosure and their institutional conditioning through digital maturity are robust to alternative profitability measures, supporting Hypotheses 1 and 2.

4.5.2. Alternative measure of cybersecurity disclosure ($CSD_{_Volume}$)

To further examine measurement sensitivity, Table 8 reports regression results using a continuous disclosure-intensity measure (CSD_Volume). This specification captures the depth and persistence of cybersecurity reporting rather than its mere existence and aligns with disclosure research indicating that textual volume reflects governance engagement and managerial attention to cyber risk (Gao et al., 2020; Radu & Smaili, 2022).

The results show that disclosure intensity is positively associated with bank profitability across both ROA and ROE models. This suggests that more extensive cybersecurity communication reflects substantive governance engagement rather than symbolic reporting behaviour. Prior evidence links detailed cyber-risk disclosure to stronger internal controls, improved monitoring, and greater organisational transparency, which collectively enhance operational performance (Calderon & Gao, 2020; Elsayed et al., 2024; Gordon et al., 2010). These findings reinforce the view that CSD operates as a governance mechanism with measurable operational consequences.

The interaction between disclosure intensity and national digital maturity remains positive and significant, indicating that detailed reporting produces stronger profitability benefits in digitally advanced institutional environments. Such environments improve regulatory monitoring, infrastructure integration, and supervisory coordination, increasing the credibility and effectiveness of cybersecurity governance (Imran et al., 2022; Laitsou et al., 2025; Privara & Caplanova, 2025).

Overall, the CSD_Volume analysis confirms that the study's core findings are robust to alternative disclosure measurement. Both disclosure presence and disclosure depth are consistently associated with stronger bank performance, particularly within digitally mature institutional settings (Bruno et al., 2023; Smaili et al., 2023).

4.6. Heterogeneity analysis by bank size

To examine whether the moderating role of national digital maturity varies across organisational scale, the sample is divided into large and small banks based on mean total assets. Separate interaction models are estimated to assess whether the performance implications of CSD differ by bank size (Mansour et al., 2024).

Table 9 reveals clear size-based heterogeneity in the interaction between CSD and digital maturity. For large banks, the interaction between CSD and DESI is positive and economically meaningful across both ROA and ROE, indicating that CSD generates stronger profitability gains in digitally mature environments. In contrast, the interaction effect is weak and statistically insignificant among small banks, suggesting that national digital maturity plays a more limited role in shaping the performance consequences of disclosure for smaller institutions.

This asymmetric pattern is theoretically consistent with governance and institutional perspectives. Large banks operate more complex digital infrastructures, face greater cyber-risk exposure, and are subject to stronger regulatory scrutiny. Consequently, CSD in large institutions is more closely integrated with internal risk-management systems, compliance processes, and board-level governance structures (Barry et al., 2022; Calderon & Gao, 2020). Digitally mature environments further reinforce these governance mechanisms through stronger supervisory coordination, infrastructure integration, and regulatory monitoring, enabling disclosure to translate more effectively into operational performance (Imran et al., 2022; Lnenicka et al., 2024).

By contrast, smaller banks typically operate under tighter resource constraints and lower public visibility, which may reduce the strategic relevance and operational integration of CSD. Even within digitally advanced countries, these structural limitations weaken the link between disclosure practices and performance outcomes (Bruno et al., 2023).

Governance controls display broadly consistent effects across size groups. Audit committee expertise, board independence, and board skills remain positively associated with profitability, although their economic influence appears stronger among large banks, reflecting the greater governance complexity associated with scale (Bruno et al., 2025; Smaili et al., 2023).

Overall, the heterogeneity analysis indicates that national digital maturity enhances the governance and performance value of CSD primarily for large banks. These findings reinforce the institutional complementarity between firm-level governance mechanisms and country-level digital readiness and provide additional support for

Table 9
Heterogeneity of the (CSD*DESI) interaction by bank size.

Variable	ROA		ROE	
	Large Banks	Small Banks	Large Banks	Small Banks
$CSD_{_Binary}$	0.192***	0.105**	0.175***	0.098*
	0.0503	0.053	0.0485	0.0524
DESI	0.265***	0.174**	0.241***	0.161*
	0.0768	0.0866	0.0751	0.0834
($CSD_{_Binary}$*DESI)	0.415***	0.108	0.372***	0.095
	0.1007	0.0885	0.0942	0.0833
AC_Expertise	0.221***	0.158**	0.194***	0.137**
	0.0592	0.0656	0.0567	0.062
BGD	0.086***	0.043	0.076***	0.036
	0.0289	0.0277	0.0285	0.0273
B_Independent	0.056***	0.022	0.049**	0.019
	0.0211	0.0204	0.0203	0.0188
B_Skills	0.077***	0.037	0.067**	0.031
	0.0284	0.0303	0.0267	0.0261
B_SIZE	−0.041–0.0218	−0.058–0.0289	−0.045–0.0262	−0.053–0.0276
B_AGE	0.192***	0.141***	0.168***	0.124**
	0.0494	0.0513	0.0472	0.049
B_LEV	−0.118–0.0286	−0.092–0.0318	−0.108–0.0281	−0.082–0.0296
Constant	0.104***	0.079**	0.091***	0.071**
	0.0302	0.0362	0.0283	0.0359
F-statistic	14.620	8.930	12.88	7.74
Prob > F	0.000	0.002	0.0	0.003
R-squared	0.289	0.172	0.271	0.158
No. of Groups	418	471	418	471
Number of Observations	4150	4710	4150	4710

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Models account for year and country effects. Banks are classified as large (above mean assets) or small (below mean).

Hypothesis 2.

4.7. Additional analysis; control for endogeneity

Because cybersecurity disclosure decisions are not randomly determined, the relationship between CSD and bank performance may be affected by self-selection, simultaneity, and observable heterogeneity. To address these concerns, this study applies three complementary identification strategies: Heckman selection correction, two-step System GMM estimation, and PSM. Each method targets a distinct econometric concern and provides triangulated robustness evidence.

The Heckman selection model confirms that disclosure decisions are non-random, as indicated by a statistically significant Inverse Mills Ratio. After correcting for selection bias, CSD remains positively associated with bank profitability, and the moderating role of DESI remains significant. System GMM further addresses simultaneity and reverse causality by using lagged internal instruments. Diagnostic tests confirm model validity, and lagged CSD continues to exert a positive effect on performance, indicating persistent governance benefits. PSM controls for observable differences between disclosing and non-disclosing banks by constructing balanced matched samples. The results remain stable, confirming that the findings are not driven by sample composition.

Across all identification strategies, two consistent patterns emerge. First, cybersecurity disclosure remains positively associated with bank profitability. Second, the performance effect of disclosure is systematically stronger in digitally mature institutional environments. These results reinforce the robustness of Hypotheses 1 and 2. Detailed estimation results and diagnostic statistics are reported in Appendix Table A1.

5. Discussion

The empirical findings reveal a consistent and robust association between CSD and bank profitability across European banking institutions. Whether measured through disclosure presence or disclosure depth, cybersecurity transparency is positively linked to financial performance. These findings suggest that CSD increasingly functions as an internal governance practice that supports operational continuity, regulatory compliance, and cyber-risk management. The results align with prior studies showing that transparent cybersecurity governance enhances organisational resilience and reduces disruption risk.

However, these findings differ from evidence reported in other institutional contexts where CSD exhibits weaker or insignificant financial effects. This divergence likely reflects structural institutional differences. European banking systems operate within relatively advanced digital infrastructure, harmonised regulatory frameworks such as DORA and NIS2, and strong supervisory enforcement (Clausmeier, 2023). Under such conditions, CSD is more likely to reflect verifiable governance implementation rather than symbolic reporting. In institutionally weaker or digitally fragmented environments, disclosure may remain less credible and therefore less capable of influencing operational performance.

A second important insight concerns the moderating role of national digital maturity. The profitability benefits associated with CSD are significantly stronger in countries with higher DESI scores. This finding supports an institutional-contingency perspective, suggesting that governance mechanisms operate within broader digital-institutional ecosystems. Digitally mature environments provide integrated infrastructure, skilled labour markets, and coordinated regulatory supervision that enhance the implementation and verification of cybersecurity governance practices. In this sense, national digital maturity does not directly generate bank profitability but shapes the conditions under which disclosure becomes operationally meaningful.

The stronger explanatory power observed when using disclosure intensity further highlights the importance of disclosure depth. Profitability improvements are primarily associated with detailed and structured cybersecurity communication rather than minimal or generic reporting. This pattern indicates that transparency generates economic value when disclosure reflects sustained governance engagement and structured cyber-risk management practices.

From a theoretical perspective, the findings primarily support agency-based governance logic by demonstrating that CSD enhances performance through operational channels, including improved monitoring efficiency, strengthened compliance discipline, and reduced disruption costs. Institutional theory is refined by showing that national digital maturity functions as a boundary condition determining when governance transparency produces measurable economic outcomes. While reputational or signaling considerations may influence disclosure behaviour, the results suggest that, in digitally mature banking environments, the economic relevance of cybersecurity transparency is driven predominantly by operational governance implementation rather than market signaling dynamics.

Overall, the findings position CSD as an institutionally embedded governance capability whose effectiveness depends on digital-institutional readiness. Importantly, these conclusions are most directly applicable to European banking systems characterised by regulatory harmonisation and advanced digital infrastructure.

6. Limitations and directions for future research

Several limitations should guide interpretation of the findings and suggest directions for future research.

First, the analysis relies on publicly disclosed cybersecurity information, which may not fully capture the sophistication of banks' internal cyber-risk management practices. Because disclosure may be selective or strategically framed, future studies could integrate supervisory incident databases, penetration-testing results, or independent cybersecurity audit ratings to improve measurement precision.

Second, although the disclosure-intensity measure provides greater granularity than binary indicators, it captures disclosure extent rather than qualitative credibility. Future research could employ advanced text-analytic approaches, including natural language processing, semantic scoring, or machine-learning classification, to distinguish substantive governance communication from boilerplate disclosure and to examine how disclosure credibility mediates performance effects.

Third, while DESI provides a robust proxy for national digital maturity, it does not capture bank-specific technological capability or cybersecurity readiness. Future multi-level research integrating macro-digital indicators with firm-level digital investment, cybersecurity spending, or technological adoption measures would improve understanding of cross-level governance interactions. Research may also examine whether organisational absorptive capacity conditions the effectiveness of cybersecurity governance within digital ecosystems.

Fourth, the focus on European banking institutions limits generalisability. Because the European context is characterised by regulatory harmonisation and relatively advanced digital infrastructure, the governance benefits of CSD may differ in institutionally fragmented or emerging banking environments. Comparative cross-regional studies could clarify whether digital maturity universally strengthens cybersecurity governance outcomes.

Finally, although multiple econometric approaches mitigate endogeneity concerns, causal inference remains constrained in observational settings. Future research could exploit regulatory shocks—such as the implementation of DORA, NIS2, or national cybersecurity directives—as quasi-natural experiments. Difference-in-differences designs, instrumental-variable strategies, or structural dynamic models could further strengthen causal identification of cybersecurity governance effects on bank performance.

7. Conclusion

This study examined how CSD and national digital maturity jointly relate to bank profitability in European banking institutions. Using a

multi-country panel dataset and a dual-measure disclosure framework, the findings show that CSD is systematically associated with stronger financial performance. The evidence indicates that cybersecurity transparency increasingly operates as a governance practice linked to operational stability, regulatory compliance, and cyber-risk management.

A central contribution of the study is demonstrating that the economic relevance of CSD depends on the surrounding digital-institutional environment. The results show that profitability differences associated with disclosure are more pronounced in digitally mature countries characterised by integrated infrastructure, coordinated regulatory supervision, and stronger cybersecurity readiness. These findings highlight the importance of institutional complementarity, indicating that governance transparency produces measurable performance outcomes when supported by enabling digital ecosystems.

The study also demonstrates that disclosure depth matters. More structured and detailed cybersecurity reporting shows stronger associations with profitability than minimal disclosure. This suggests that transparency contributes to financial outcomes when disclosure reflects substantive governance engagement rather than symbolic communication.

From a theoretical standpoint, the findings provide evidence that CSD operates primarily through operational governance channels consistent with agency-based explanations. The results also indicate that national digital maturity shapes the effectiveness of governance transparency across institutional environments. These theoretical implications are most directly relevant to European banking systems characterised by relatively advanced regulatory coordination and digital development.

The findings offer practical implications for financial institutions and policymakers. Banks may benefit from integrating CSD with substantive governance and risk-management practices rather than treating transparency solely as a compliance requirement. Policymakers may strengthen the effectiveness of cybersecurity governance by supporting digital infrastructure development, regulatory coordination, and supervisory capacity.

As digital transformation continues to reshape financial systems, CSD, when embedded within mature digital and regulatory environments, is likely to remain an important governance practice associated with organisational resilience and sustainable bank performance.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used **QuillBot** for assistance in proofreading and enhancing the manuscript's readability. Following the use of this tool, the author(s) accurately reviewed and revised the content as necessary. The author(s) fully assume responsibility for the content of the publication.

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CRediT authorship contribution statement

Marwan Mansour: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Ala Hussein Albawwat:** Writing – review & editing, Visualization, Project administration, Methodology, Investigation. **Ahmad Marei:** Writing – review & editing, Supervision, Software, Resources, Project administration, Funding acquisition. **Hady O. Abozeid:** Writing – review & editing, Investigation, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Marwan Mansour

Appendix

Table A1.

Table A1
Robustness check: endogeneity control models.

Variables	Model 1 ROA	Model 2	Model 1 ROE	Model 2	Model 1 ROA	Model 2	Model 1 ROE	Model 2	Model 1 ROA	Model 2	Model 1 ROE	Model 2
	Heckman		Heckman		GMM		GMM		PSM		PSM	
CSD t-1	-	-	-	-	0.263***	0.273***	0.224***	0.236***	-	-	-	-
					(3.62)	(3.76)	(3.29)	(3.42)				
CSD	0.287***	0.291***	0.232***	0.248***	0.193***	0.204***	0.179***	0.191***	0.172***	0.189***	0.166**	0.176**
	(3.71)	(3.85)	(3.18)	(3.31)	(3.17)	(3.25)	(2.96)	(3.07)	(2.82)	(2.91)	(2.41)	(2.49)
DESI	0.168**	0.174**	0.149**	0.152**	0.204***	0.217***	0.213***	0.228***	0.138**	0.148**	0.144**	0.153**
	(2.09)	(2.18)	(2.27)	(2.33)	(3.33)	(3.41)	(3.82)	(3.92)	(2.14)	(2.22)	(2.03)	(2.18)
INRERACTION	-	**0.401*** **	-	**0.355*** **	-	0.428***	-	0.387***	-	0.376**	-	0.312**
(CSD × DESI)		**(3.81)**		**(3.47)**		(4.22)		(3.96)		(2.27)		(2.09)
AC_Expertise	0.221***	0.231***	0.163**	0.174**	0.367***	0.387***	0.426***	0.443***	0.245***	0.263***	0.083**	0.091**
	(3.41)	(3.56)	(2.65)	(2.85)	(3.82)	(3.93)	(3.98)	(4.12)	(3.63)	(3.81)	(2.54)	(2.64)
BGD	0.071**	0.078**	0.062**	0.069**	0.084***	0.093***	0.078**	0.087**	0.134***	0.145***	0.053*	0.058*
	(2.43)	(2.51)	(2.23)	(2.34)	(2.91)	(3.02)	(2.73)	(2.85)	(3.41)	(3.57)	(1.78)	(1.91)
B_Independent	0.042**	0.045**	0.035**	0.039**	0.043***	0.048***	0.039**	0.042**	0.023*	0.027*	0.031**	0.033**
	(2.05)	(2.12)	(1.94)	(2.05)	(3.51)	(3.61)	(2.61)	(2.77)	(1.77)	(1.87)	(2.05)	(2.12)
B_Skills	0.051**	0.058**	0.045**	0.051**	0.161***	0.174***	0.138***	0.145***	0.029**	0.033**	0.026*	0.029*
	(2.33)	(2.45)	(2.02)	(2.16)	(3.64)	(3.82)	(3.62)	(3.71)	(2.27)	(2.41)	(1.82)	(1.94)
B_SIZE	−0.032	−0.035	−0.029	−0.031	−0.045**	−0.048**	−0.031*	−0.033*	−0.413***	−0.433***	−0.952***	−0.984***
	(1.65)	(1.74)	(1.61)	(1.68)	(2.55)	(2.61)	(1.96)	(2.03)	(5.61)	(5.73)	(5.82)	(5.97)
B_AGE	0.218***	0.225***	0.194***	0.201***	0.493***	0.518***	0.412***	0.426***	0.456***	0.474***	0.271***	0.282***
	(3.98)	(4.12)	(3.88)	(4.01)	(5.01)	(5.18)	(4.56)	(4.66)	(6.02)	(6.12)	(10.62)	(10.82)

(*continued on next page*)

Table A1 (*continued*)

Variables	Model 1 ROA	Model 2	Model 1 ROE	Model 2	Model 1 ROA	Model 2	Model 1 ROE	Model 2	Model 1 ROA	Model 2	Model 1 ROE	Model 2
	Heckman		Heckman		GMM		GMM		PSM		PSM	
B_LEV	−0.082**	−0.089**	−0.125***	−0.132***	−0.049**	−0.054**	−0.091***	−0.098***	−0.124***	−0.134***	−0.233***	−0.243***
	(2.83)	(2.91)	(3.85)	(3.94)	(2.66)	(2.77)	(4.71)	(4.86)	(6.61)	(6.79)	(11.92)	(12.11)
IMR (lambda)	0.059***	0.063***	0.046**	0.049**	-	-	-	-	-	-	-	-
	(2.66)	(2.79)	(2.23)	(2.35)								
Constant	0.089**	0.093**	0.209***	0.217***	0.066**	0.071**	0.022*	0.025*	−0.081*	−0.087*	0.129***	0.138***
	(2.57)	(2.65)	(3.35)	(3.45)	(2.65)	(2.79)	(1.84)	(1.94)	(1.69)	(1.78)	(4.77)	(4.97)
Hansen J. (p-value)	-	-	-	-	0.231	0.217	0.267	0.281	-	-	-	-
AR (1)	-	-	-	-	0.002	0.001	0.003	0.002	-	-	-	-
AR (2)	-	-	-	-	0.342	0.321	0.419	0.408	-	-	-	-
Pseudo-R2	0.298	0.312	0.273	0.289	-	-	-	-	0.251	0.267	0.222	0.238
NO. of groups	889	889	889	889	889	889	889	889	889	889	889	889
Number of obs.	8890	8890	8890	8890	8890	8890	8890	8890	8890	8890	8890	8890

Note: Robust t-statistics are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Models include year and country fixed effects. Model 2 includes the interaction term (CSD × DESI). IMR (lambda) is specific to Heckman models.

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